Evaluating Temporal Link Prediction methods to generate synthetic social networks data

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Problem Statement

- Social networks: critical for understanding modern online communities
- Risky users exploit platforms for malicious activities
- **Goal**: Temporal link prediction for social network analysis

Objectives

1. Predict future user connections
2. Evaluate incremental learning
3. Generate synthetic networks

Dataset: Gab Social Network

- Platform: Gab (2016–2025)
- 6 temporal snapshots
- Posts + social network graphs
- BERT embeddings provided

Property	Value
Total Posts	182,545
Unique Users	27,002
Snapshots	6
Time Span	2016–2025

Synthetic Dataset

Synthetic Users

- 3 batches: $400 + 400 + 200$
- Total: 1,000 synthetic users
- 9,990 synthetic posts
- Pre-computed BERT embeddings

Purpose

- Test model generalization
- Generate synthetic network links
- Evaluate structural realism

Pipeline: Real graph + synthetic embeddings → trained model → predicted edges

Data Quality: Missing Values

Attribute	Missing %
Language	0.48%
All others	0.00%

- 99.52% overall completeness
- No empty content posts
- 100% snapshot consistency

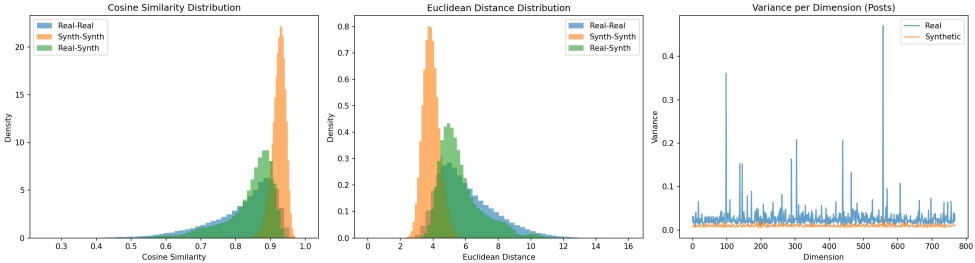
Snapshot	Snap.	Incr.	Cov.
2016-2021	18,926	18,926	100%
2022	13,108	32,034	100%
2023	13,376	45,410	100%
2024	13,400	58,810	100%
Jan-Jul 25	44,878	103,688	100%
Jul 25	78,857	182,545	100%

Duplicate Content Analysis

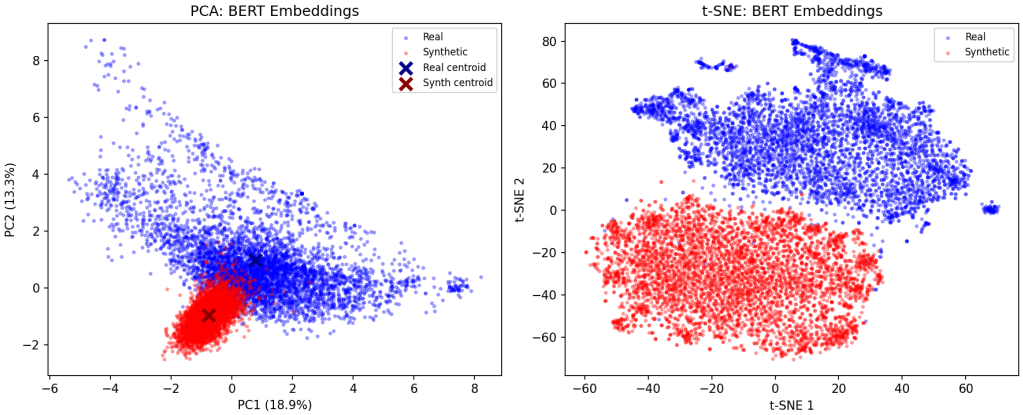
Snapshot	Dup. %
2016-2021	1.79%
2022	3.02%
2023	4.21%
2024	4.20%
Jan-Jul 25	7.52%
Jul 25	10.04%

- Increasing duplicate rate over time
- No duplicate post IDs
- Each post has unique metadata

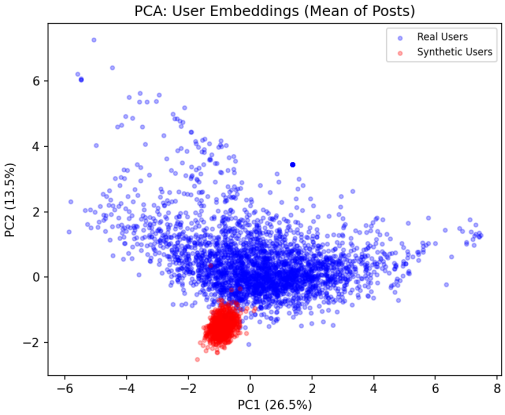
BERT Embeddings: Real vs Synthetic

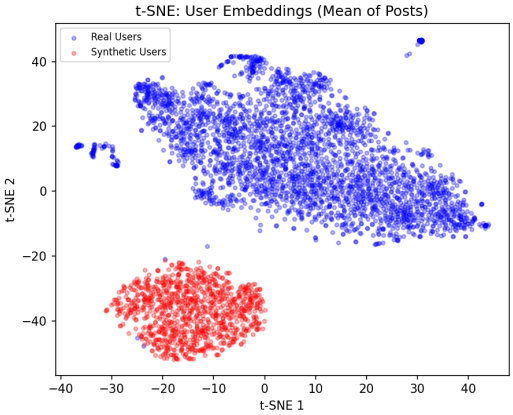


Embedding Space Visualization



User-Level Embedding Analysis





Embedding Statistics

Metric	Real	Synthetic
Posts	182,545	9,990
Cos. Sim. (mean)	0.821	0.924
Cos. Sim. (std)	0.097	0.019
Variance	0.026	0.010

Key Finding

Synthetic users: variance ratio 11.23× lower than real users

Feature Construction

Node features from BERT embeddings:

$$\mathbf{h}_u^{(t)} = \frac{1}{|P_u^{(t)}|} \sum_{p \in P_u^{(t)}} \text{BERT}(\text{content}_p)$$

- 768-dimensional feature vectors
- Temporal feature construction
- No data leakage

Graph Structure

- Directed graph $G_t = (V_t, E_t)$
- Cumulative edge assumption
- Sparse adjacency representation

Adjacency Matrix Normalization

$$\hat{A}_t = \tilde{D}_t^{-\frac{1}{2}} \tilde{A}_t \tilde{D}_t^{-\frac{1}{2}}, \quad \tilde{A}_t = A_t + I$$

Training Input:

- Historical adjacency list $[\hat{A}_{t-k}, \ldots, \hat{A}_{t-1}]$
- Historical node features $[H_{t-k}, \ldots, H_{t-1}]$
- Node masks for active nodes

Training Output:

- Edge pairs (u, v)
- Binary labels $y \in \{0, 1\}$
- Negative sampling

EvolveGCN Architecture

Key Innovation

RNN evolves GCN parameters instead of node embeddings

Graph Convolution:

$$H_t^{(l+1)} = \sigma(\hat{A}_t H_t^{(l)} W_t^{(l)})$$

Weight Evolution (EvolveGCN-H):

$$W_t^{(l)} = \mathrm{GRU}(H_t^{(l)}, W_{t-1}^{(l)})$$

- Handles new nodes at test time
- No requirement for static node sets
- Captures temporal dynamics

GCN Architecture Details

GRCU Layers:

- Layer 1: 768 $\rightarrow$ 128
- Layer 2: 128 $\rightarrow$ 64

MLP Classifier:

- Input: 128 (64 $\times$ 2 concatenated)
- Output: 2 logits

Loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_i \log(\hat{y}_i)$$

Top-K Selection

$$\text{scores} = \frac{H_t^{(l)} \cdot s}{\|s\|}$$

Select most relevant embeddings for weight evolution

Training Pipelines

Original Implementation

- Sequence-based training
- All snapshots in single batch
- 50 epochs total
- GRU manages temporal flow

Incremental Implementation

- Phase-based training
- One snapshot per phase
- 50 epochs per phase
- Weight transfer between phases

Configurations: LR $\in \{0.001, 0.0005\}$, NM $\in \{1, 2\}$

Classification Results

	LR	NM	MAP	AUC	Prec.	Rec.	F1
Incr.	0.001	1	0.78	0.78	0.25	0.49	0.33
	0.001	2	0.64	0.74	0.75	0.68	0.69
	0.0005	1	0.78	0.80	0.68	0.52	0.38
	0.0005	2	0.62	0.72	0.73	0.68	0.69
Orig.	0.001	1	**0.87**	0.63	**0.86**	**0.86**	**0.86**
	0.001	2	0.77	0.65	0.82	0.83	0.83
	0.0005	1	0.86	0.64	0.82	0.79	0.78
	0.0005	2	0.76	0.65	0.81	0.84	0.81

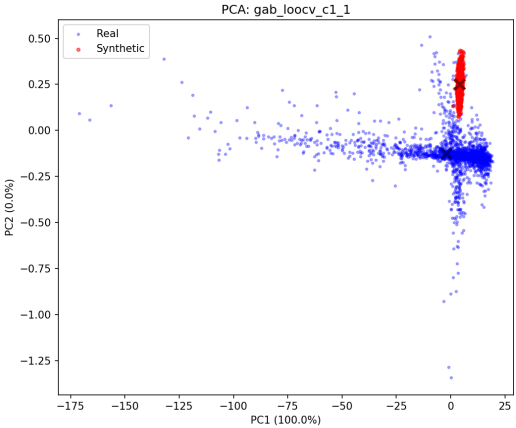
Incr. = Incremental, Orig. = Original, LR = Learning Rate, NM = Negative Multiplier

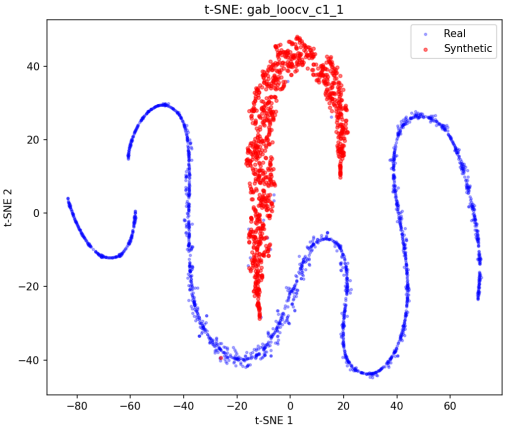
Per-Class Metrics

	LR	NM	Class 0 (Non-link)			Class 1 (Link)		
			P	R	F1	P	R	F1
Incr.	0.001	1	0.00	0.00	0.00	0.49	0.97	0.65
	0.001	2	0.77	0.92	0.83	0.73	0.44	0.55
	0.0005	1	0.85	0.04	0.08	0.51	0.99	0.67
	0.0005	2	0.77	0.90	0.83	0.69	0.47	0.56
Orig.	0.001	1	0.89	0.83	0.86	0.84	0.89	0.87
	0.001	2	0.89	0.88	0.88	0.76	0.78	0.77
	0.0005	1	0.93	0.62	0.74	0.71	0.96	0.82
	0.0005	2	0.95	0.76	0.84	0.66	0.93	0.77

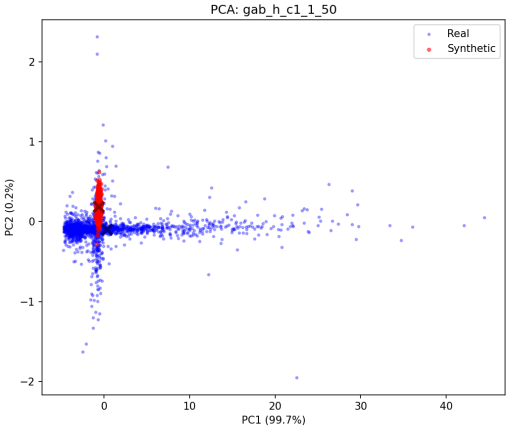
- Original: balanced class prediction across all configs
- Incremental (NM=1): severe class imbalance, predicts mostly Class 1

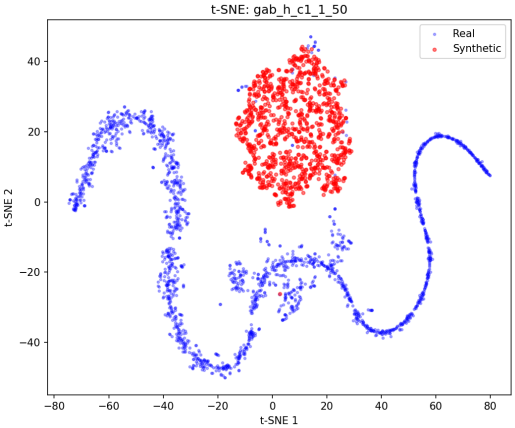
GCN Embeddings: Original Implementation



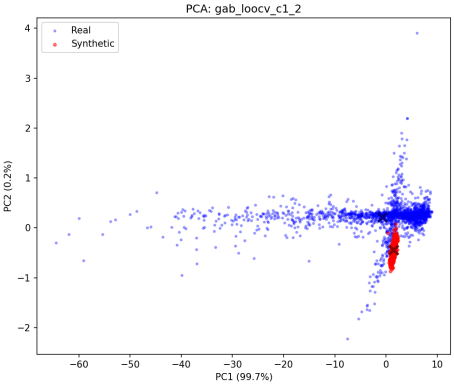


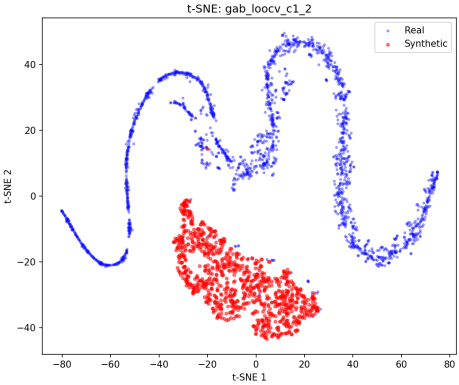
GCN Embeddings: Incremental Implementation



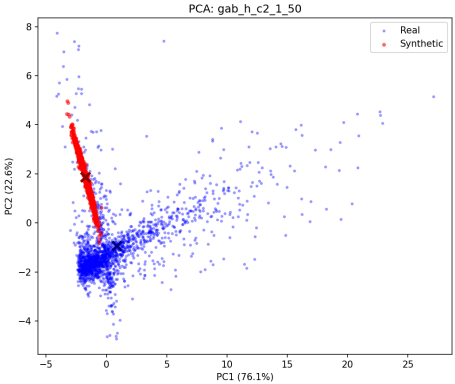


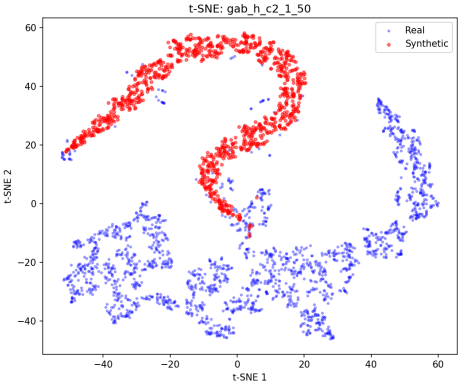
All Configurations: Original





All Configurations: Incremental





PCA Statistics

	Config	PC1	PC2	Total	Centroid
Orig.	LR=0.001, NM=1	99.98%	0.01%	100.00%	6.29
	LR=0.001, NM=2	99.71%	0.23%	99.94%	2.33
	LR=0.0005, NM=1	95.99%	3.39%	99.38%	2.06
	LR=0.0005, NM=2	99.96%	0.03%	100.00%	6.65
Incr.	LR=0.001, NM=1	99.65%	0.21%	99.86%	1.01
	LR=0.001, NM=2	99.39%	0.57%	99.96%	3.41
	LR=0.0005, NM=1	76.13%	22.61%	98.74%	3.84
	LR=0.0005, NM=2	91.20%	8.47%	99.67%	4.78

Key Finding

Original: Dimensional collapse (PC1 $\approx$ 96-100%)
Incremental (LR=0.0005, NM=1): Richer embedding space (PC1 = 76%)

Graph Metrics Comparison

Network	Avg Deg	Mod.	Avg Path	Clust.
Facebook	43.69	0.83	3.62	0.61
Twitter	21.75	0.80	4.90	0.57
Google+	127.06	0.47	3.32	0.49
Gab (Ours)	32.11	0.27	3.04	0.08
Synthetic (Incr. LR=.0005 NM=1)	220.35	-0.001	1.02	0.59

- Synthetic: over-dense, no community structure (modularity ≈ 0)
- Gab: lower clustering than mainstream networks

Key Observations

Original Implementation

- Best MAP: 0.87
- Best F1: 0.86
- Severe oversmoothing
- No synthetic connections

Incremental Implementation

- Best AUC: 0.80
- Richer embeddings
- Attempts synthetic links
- Poor structural fidelity

Achievement of Goals

Achieved

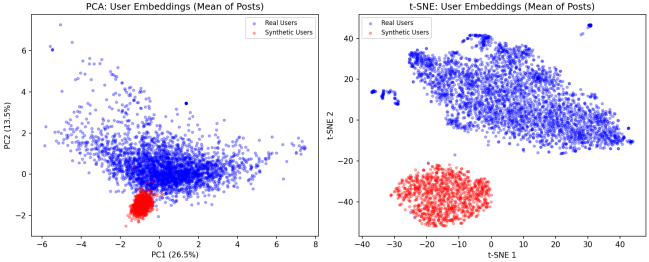
- Temporal link prediction model
- Quantitative evaluation
- Embedding space analysis

Failed

- Realistic synthetic networks
- Generalization to synthetic users

Limitations

1. **Embedding Mismatch**: Synthetic users too homogeneous (variance ratio 11.23×)
2. **Oversmoothing**: Dimensional collapse in GCN layers
3. **Cumulative Edge Assumption**: Does not reflect unfollowing behavior



Future Work

1. **Mitigate Oversmoothing**
   - Jumping knowledge networks
   - APPNP-style propagation
   - Fewer GCN layers
2. **Improve Synthetic Data**
   - Diverse generation prompts
   - Embedding perturbation
3. **Alternative Models**
   - Other temporal GNN architectures
   - Attention-based approaches

Thank You 🚀

Questions?